Instituto Politécnico Nacional

Escuela Superior De Cómputo

Genetic Algorithms

Presentación

Cuckoo Search Via Lévy Flights

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GENETIC ALGORITHMS

CUCKOO SEARCH VIA LÉVY FLIGHTS

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INTRODUCTION

Cuckoo Search is a optimization algorithm. It reimagines how cuckoo birds solve their reproductive challenges as a method for solving complex computational problems.

Origin Story

- Created in 2009 by researchers Xin-She Yang and Suash Deb
- Born from observing nature's ingenious problem-solving mechanisms
- Part of a growing field of bio-inspired computational intelligence

BIRD CUCKOO

Cuckoos are birds and in the Cuculidae family. The family Cuculidae contains 150 species, which are divided into 33 genera.

- Core Inspiration: Reproductive behavior of cuckoo birds
- Key Characteristics:
  - Mimics cuckoo's egg-laying strategy
  - Utilizes Lévy flight mechanism
  - Probabilistic optimization technique

THE CUCKOO'S SURVIVAL STRATEGY

- Some cuckoo species are brood parasites
- Unique Reproduction Mechanism:

1. Lay eggs in nests of other bird species
2. Eggs are camouflaged to resemble host bird's eggs
3. Host birds unknowingly incubate and raise cuckoo chicks

WHAT IF EGG IS DISCOVERED?

- Discovered foreign egg will be thrown or host will leave nest.
- Nests with eggs are selected.
- Cuckoo eggs will hatch earlier than host egg.

THEN WHAT?

- Cuckoo chick will evict all host eggs.
- Increased food share.

Egg Mimicry:

- Cuckoo eggs closely resemble host eggs
- Reduces chances of egg rejection

Host Selection:

- Carefully choose nests with similar egg appearances
- Maximize offspring survival probability

VIDEO


1 MINUTE ANIMALS
WHEN IT COMES TO THEIR
PARENTAL PRACTICES
@kungfubitd

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LÉVY FLIGHTS

- A random walk with a probability distribution of Lévy
- Characterized by long, unpredictable jump lengths
- Food search in nature is random or quasi-random.
- Foraging path is random walk and depends on current location and transition probability.
- Since next direction is based on probability, it can be modeled mathematically.

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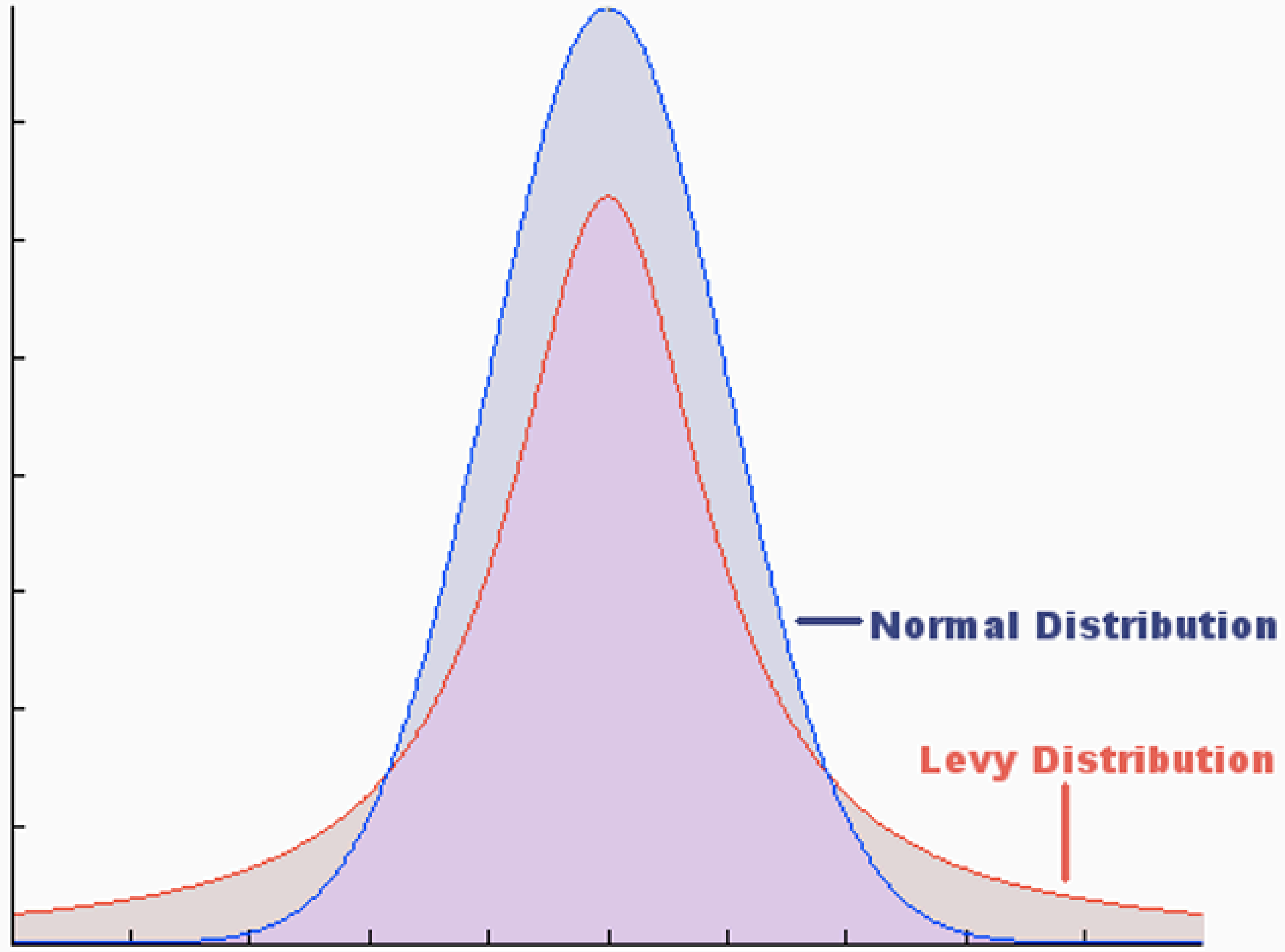
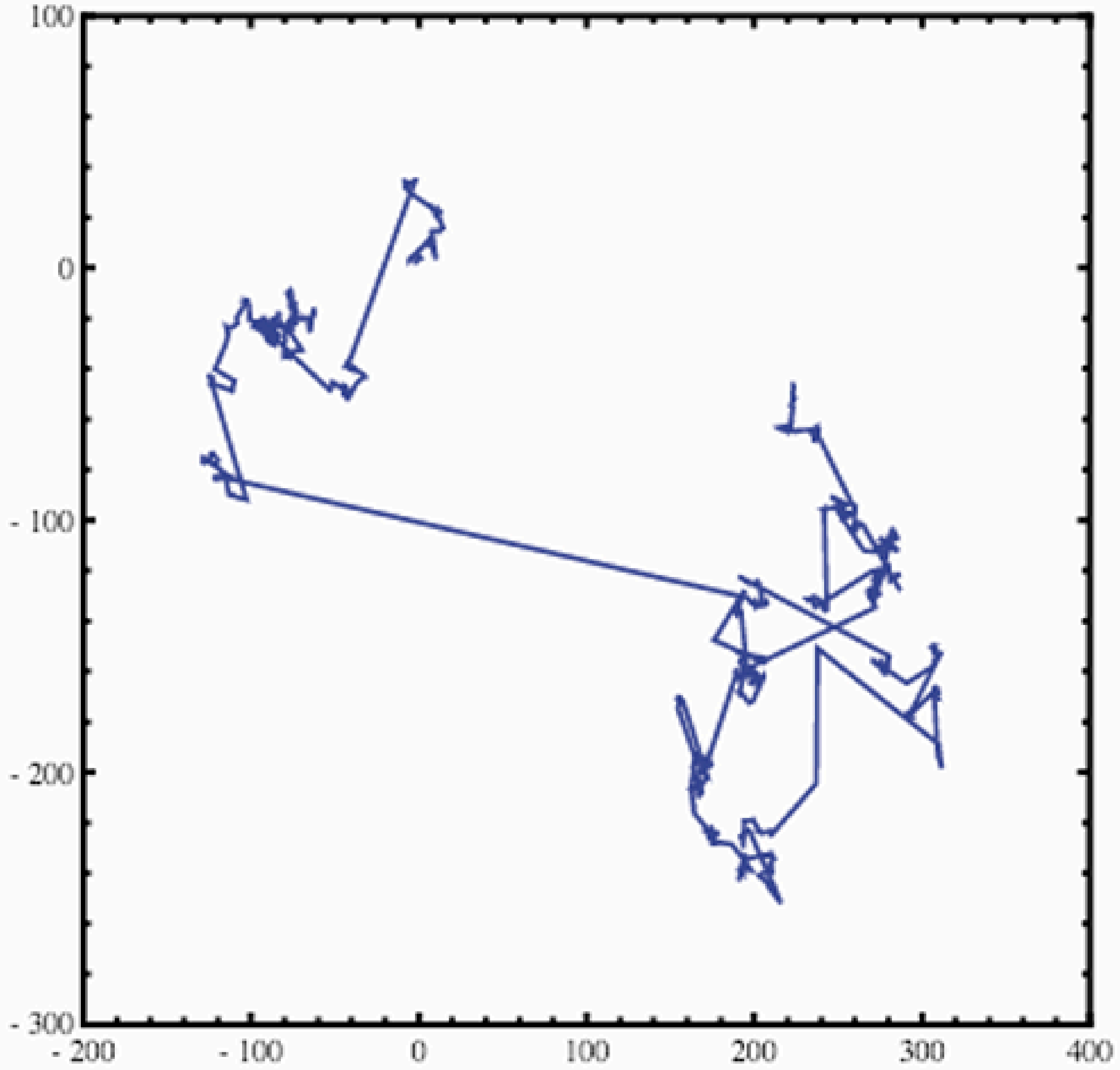
DIFFERENCE FROM RANDOM WALK



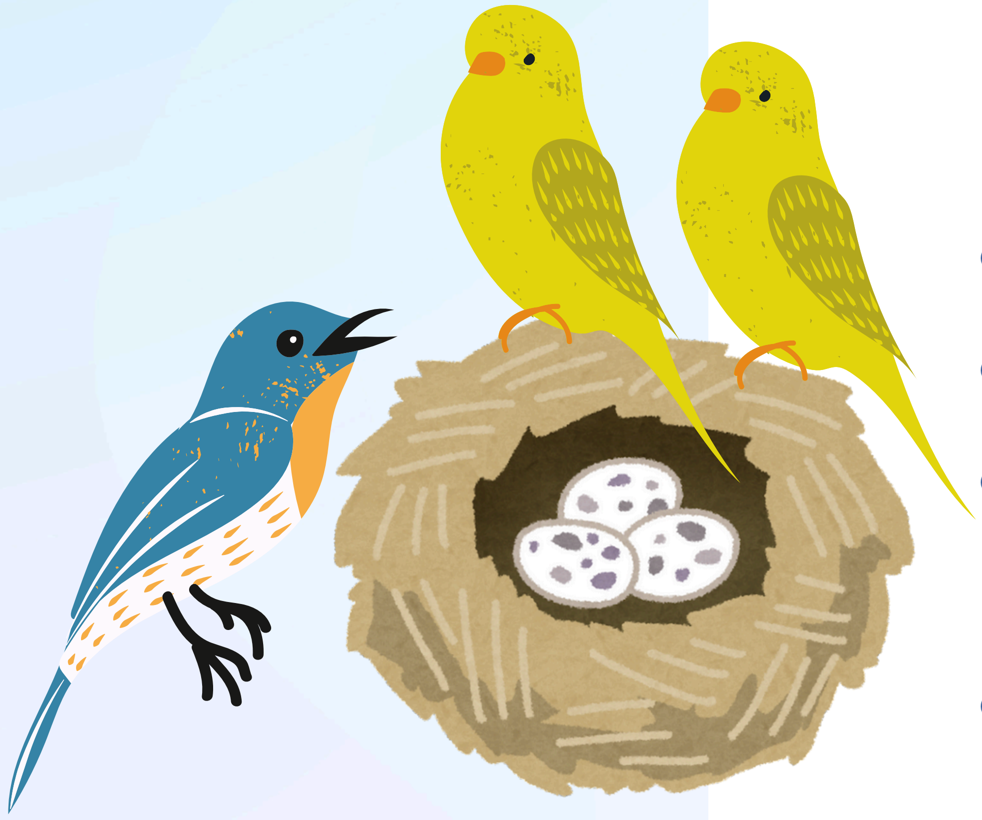


Wikipedia

BIOLOGICAL INSPIRATION

- Eggs in nests : set of solutions
- Cuckoo egg : new solution.
- New and better solutions will replace, less fit solutions.
- Cuckoo's change position with Lévy flights after leaving nest.

RULES OF IMPLEMENTATION



- Each cuckoo can lay one egg at each time step.
- High quality nests will carry onto next generations.
- # of host nests is fixed and pa is the probability of discovery of an alien egg.
- Host bird can throw away egg or leave nest.

PSEUDO CODE

Cuckoo Search via Lévy Flights

```
begin
  Objective function f(x),   x = (x_1, ..., x_d)^T
  Generate initial population of
      n host nests x_i (i = 1, 2, ..., n)
  while (t <MaxGeneration) or (stop criterion)
    Get a cuckoo randomly by Lévy flights
      evaluate its quality/fitness F_i
    Choose a nest among n (say, j) randomly
    if  (F_i > F_j),
        replace j by the new solution;
    end
    A fraction (p_a) of worse nests
        are abandoned and new ones are built;
    Keep the best solutions
        (or nests with quality solutions);
    Rank the solutions and find the current best
  end while
  Postprocess results and visualization
end
```

PSEUDO CODE

Initialization

Parameters

- n : number of host nests
- pa : probability of discovery of alien egg
- MaxGeneration : maximum number of iterations

Initialization

Generate initial n host, xi( )

Evaluate f(xi( ))

Cuckoo Search via Lévy Flights

begin
  Objective function $f(\mathbf{x})$, $\mathbf{x} = (x_1, ..., x_d)^T$
  Generate initial population of
    n *host nests* $\mathbf{x}_i$ $(i = 1, 2, ..., n)$
  while $(t <$*MaxGeneration*$)$ *or (stop criterion)*
    Get a cuckoo randomly by Lévy flights
      evaluate its quality/fitness F_i
    Choose a nest among n *(say,* j*) randomly*
    if $(F_i > F_j)$,
      replace j *by the new solution;*
    end
    A fraction (p_a) *of worse nests*
      are abandoned and new ones are built;
    Keep the best solutions
      (or nests with quality solutions);
    Rank the solutions and find the current best
  end while
  Postprocess results and visualization
end

PSEUDO CODE

- Generate a new solution

$$x_i^{(t+1)} = x_i^{(t)} + \alpha \oplus Le'vy(\lambda)$$

  - Evaluate

$$f(x_i^{(t+1)})$$

- Choose a nest xj randomly

If $f(x_j^{(t)}) < f(x_i^{(t+1)})$

Replace $x_j^{(t)}$ with $x_i^{(t+1)}$

- Abandon a fraction of pa worse nests.
  - Build new nests with Lévy flights
  - Keep the best solutions

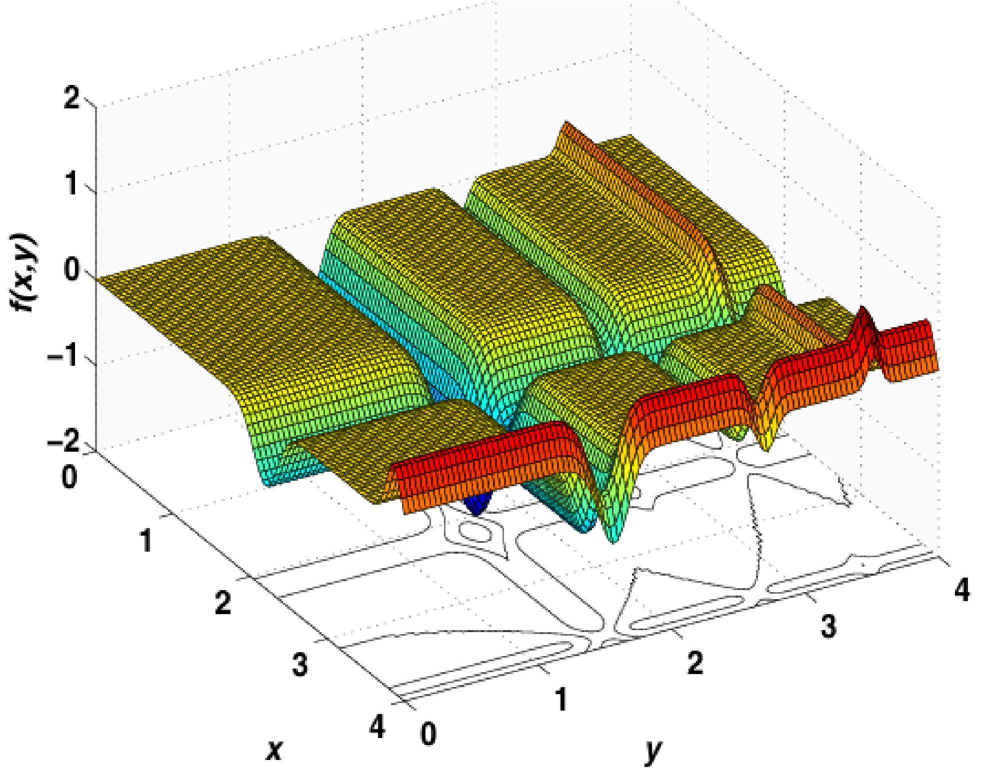
Cuckoo Search via Lévy Flights

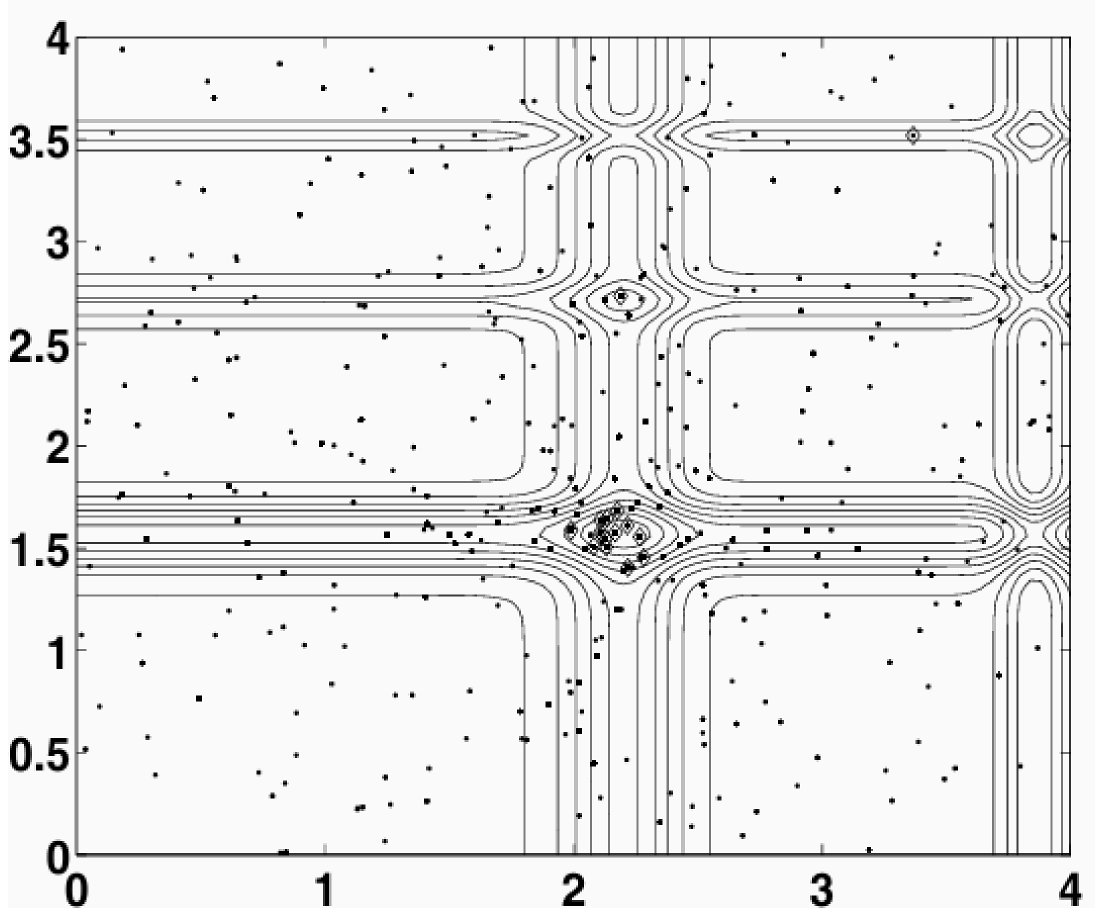
begin
 Objective function $f(\mathbf{x}),\ \ \mathbf{x} = (x_1, ..., x_d)^T$
 Generate initial population of
  n host nests $\mathbf{x}_i\ (i = 1, 2, ..., n)$
 while *(t <MaxGeneration) or (stop criterion)*
  Get a cuckoo randomly by Lévy flights
   evaluate its quality/fitness F_i
  Choose a nest among n (say, j) randomly
  if $(F_i > F_j)$,
   replace j by the new solution;
  end
  A fraction (p_a) of worse nests
   are abandoned and new ones are built;
  Keep the best solutions
   (or nests with quality solutions);
  Rank the solutions and find the current best
 end while
 Postprocess results and visualization
end

REALISATION AND VERIFICATION

Bivariate Michaelwicz function

$$f(x,y) = -\sin(x)\, sin^{2m}\left(\frac{x^2}{\pi}\right) - \sin(y)\, sin^{2m}\left(\frac{2y^2}{\pi}\right)$$





Search paths of nests using Cuckoo Search.The final locations of the nests are marked with ◇ in the figure.

COMPARISON OF CS WITH GENETIC ALGORITHMS

We can see that the CS is much more efficient in finding the global optima with higher success rates. Each function evaluation is virtually instantaneous on modern personal computer.

Functions/Algorithms	GA	CS
Multiple peaks	$52124 \pm 3277(98\%)$	$927 \pm 105(100\%)$
Michalewicz's (d=16)	$89325 \pm 7914(95\%)$	$3221 \pm 519(100\%)$
Rosenbrock's (d=16)	$55723 \pm 8901(90\%)$	$5923 \pm 1937(100\%)$
De Jong's (d=256)	$25412 \pm 1237(100\%)$	$4971 \pm 754(100\%)$
Schwefel's (d=128)	$227329 \pm 7572(95\%)$	$8829 \pm 625(100\%)$
Ackley's (d=128)	$32720 \pm 3327(90\%)$	$4936 \pm 903(100\%)$
Rastrigin's	$110523 \pm 5199(77\%)$	$10354 \pm 3755(100\%)$
Easom's	$19239 \pm 3307(92\%)$	$6751 \pm 1902(100\%)$
Griewank's	$70925 \pm 7652(90\%)$	$10912 \pm 4050(100\%)$
Shubert's (18 minima)	$54077 \pm 4997(89\%)$	$9770 \pm 3592(100\%)$

COMPARISON OF CS WITH PARTICLES WARM OPTIMISATION

For all the test functions, CS has outperformed both GA and PSO. The primary reasons are two folds: A fine balance of randomization and intensification, and less number of control parameters.

TABLE II
COMPARISON OF CS WITH PARTICLE SWARM OPTIMISATION

Functions/Algorithms	PSO	CS
Multiple peaks	$3719 \pm 205(97\%)$	$927 \pm 105(100\%)$
Michalewicz's (d=16)	$6922 \pm 537(98\%)$	$3221 \pm 519(100\%)$
Rosenbrock's (d=16)	$32756 \pm 5325(98\%)$	$5923 \pm 1937(100\%)$
De Jong's (d=256)	$17040 \pm 1123(100\%)$	$4971 \pm 754(100\%)$
Schwefel's (d=128)	$14522 \pm 1275(97\%)$	$8829 \pm 625(100\%)$
Ackley's (d=128)	$23407 \pm 4325(92\%)$	$4936 \pm 903(100\%)$
Rastrigin's	$79491 \pm 3715(90\%)$	$10354 \pm 3755(100\%)$
Easom's	$17273 \pm 2929(90\%)$	$6751 \pm 1902(100\%)$
Griewank's	$55970 \pm 4223(92\%)$	$10912 \pm 4050(100\%)$
Shubert's (18 minima)	$23992 \pm 3755(92\%)$	$9770 \pm 3592(100\%)$

ADVANTAGES AND DISADVANTAGES

- High global exploration capability
- Fast Convergence
- Easy deployment
- Few parameters to adjust

- Performance may vary depending on the issue
- Sensitivity to initial parameter selection
- You can get stuck in local optimums

APPLICATIONS

- Engineering optimization problems
- NP-hard combinatorial optimization problems
- Data fusion in wireless sensor networks
- Neural network training
- Manufacturing scheduling
- Nurse scheduling

CONCLUSION

We have formulated a new metaheuristic Cuckoo Search in combination with Lévy flights, based on the breeding strategy of some cuckoo species. The proposed algorithm has been validated and compared with other algorithms such as genetic algorithms and particles warm optimization. Simulations and comparison show that CS is superior to these existing algorithms for multimodal objective functions. This is partly due to the fact that there are fewer parameters to be fine-tuned in CS than in PSO and genetic algorithms.

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THANK YOU

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